FG.02.07.01 FIDM Interface

Project HMI Subsystems 1

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Configuration FG.02.07 Function Fully Integrated Display Module MY 23 Cadence 160

Configuration Type Global Configuration

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1 FIDM Interface Subsystem Requirements

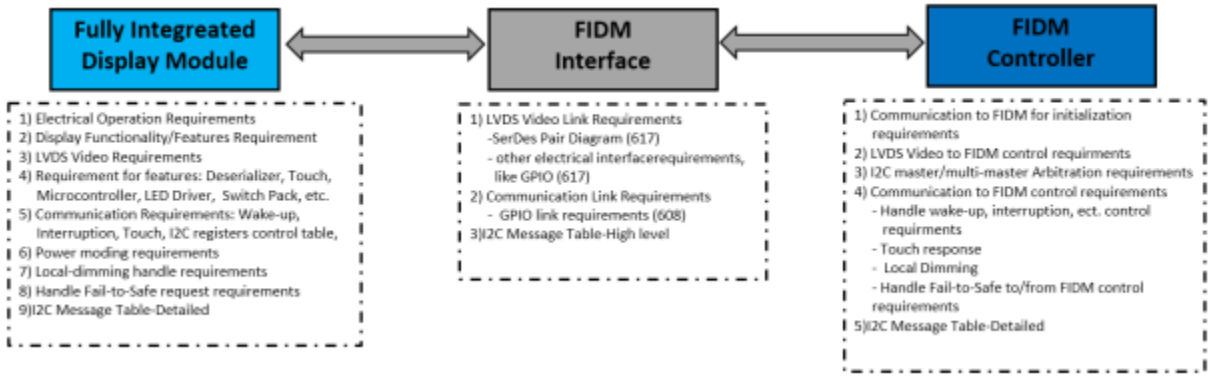
1.1 Subsystem Description

This module describes the interface between the FG.02.07.02 FIDM Controller and the FG.02.07.03 Fully Integrated Display Module functions. This module is intended to document the information that is common to both the FIDM and FIDM Controller such as the master message list.

1.1.1 Interface Summary

The interface routing required for the specific application need be implemented. Video source FIDM Controller (FG.02.07.02) and video sink FIDM (FG.02.07.03) need to initialize and maintain the indicated SerDes configuration to support the functions needed. Video and controls signals will be transported over high speed FPD Link cable. The I2C and interrupt communication interface is transported over the video LVDS backchannel. FIDM Controller (FG.02.07.02) will be the I2C master and FIDM (FG.02.07.03) will be I2C slaves. If touch over I2C is implemented, an interrupt will be used to indicate additional information, is available at the touch controller. If FIDM based device (i.e. AUX, ICS, HVAC, HUD, IPC, etc.) controls are implemented over I2C, an additional interrupt will indicate additional information is available at the device. The touch controller interrupt will be active low and other controls interrupt will be active high.

FIDM Interface: The message map table to interface between FIDM (FG.02.07.03) and FIDM-Controller (FG.02.07.02)



1.1.2 Module Revision Log

1.1.2.1 CR 2603325 April 2020 Changes

Defined explicit periodic rate handling for Touch Controller diagnostic messages with a slower background rate when no failures are detected in order to reduce static interrupt/message loading on the FIDM controller.

Reduced periodic rate when no fault present for Display Knob diagnostic to reduce static interrupt/message loading on FIDM Controller.

1.1.2.2 CR 2875263 Mar-2021 Changes for Physical Switch Diagnostics in Cadence 160

1.1.2.2.1 Interface defined for Front HVAC Controls for FIDM HVACSwitch pack, Sec1.2

Includes I2C interfaces for remote physical switch/knob back reads, diagnostic read for

forwarding to the BCM as new CAN signals, and switch indicator (embedded LEDs within switch buttons driven by FIDM micro) writes.

1.1.2.2.2 Privacy Mode control for FPI DIsplays only(N/A to VCU), Sec 1.2

This is a signal I2C write to be able to turn On/Off the privacy mode polarizer in a front passenger Infotainment display.

1.2 FIDM microprocessor <-> FIDM Controller I2C Messages

1.2.1 Message Table

The table1 below defines the master list of I2C messages between the FIDM microprocessor and the FIDM Controller. This table does not include messages between the internal FIDM touch-controller and its associated FIDM Controller driver.

For Data Read→ Provider is FIDM & Consumer is FIDM Controller. For Data Write →Provider is FIDM Controller & Consumer is FIDM.

Data Address	Data Length (Bytes)	Data	Data Direction (From Device)	Data Type	Update Rate	Feature Implemented in ICS Controls or Reserved for Future Use
0x00	2	Hardware Version	Read	FIDM Hardware Version. The specific value shall be defined in the Display CTS. FIDM supplier is responsible & shall increment this data every time when there is major change in the electrical schematic. Provider is FIDM & consumer is FIDM-Controller. Note- FIDM Supplier	On need from master.	Display Unit controls

				should be within (0x0000-0x7FFF)		
0x02	2	Firmware Version	Read	FIDM Firmware Version. The specific value shall be defined in the Display CTS. FIDM supplier is responsible & shall increment this data every time when there is Software related change in FIDM module. Provider is FIDM & consumer is FIDM-Controller.	On need from master.	Display Unit controls
0x04	1	Touch Controller ID The Display Master should use the standard TC driver from Touch vendor	Read	Identification number of touch controller used. The specific value shall be defined in the Display CTS. Provider is FIDM & consumer is FIDM-Controller. FIDM supplier is responsible for this data to indicate specific Touch controller	On need from master.	Display Unit controls

				used in FIDM module.		
0x05	1	Display ID	Read	Identification number of display resolution / timing used. The specific value shall be defined in DNG-FIDM Interface section 1.3. Provider is FIDM & consumer is FIDM-Controller. FIDM supplier is responsible & shall use this data to indicate specific Display variant incorporated within FIDM module.	On need from master.	Display Unit controls
0x06	1	HVAC switch pack via FIDM Present	Read	Identification to determine present or absent True = 0x01 →Present False = 0x00 → Absent	On need from master.	Display Unit controls
0x08	1	Knob1 Counts Change(Volume)	Read	Mechanical Technology: An eight bit	Due to IRQ or need from	Display Unit controls

				signed integer value representing the changes in encoder counts which have been accumulated since the last time a change in the encoder was read. Clockwise is positive. while count overflow: max value shall be 127 for CW and -128 for CCW respectively Provider is FIDM & consumer is FIDM-Controller.	master.	
0x09	1	Knob2 Counts Change(Tune/Select)	Read	Mechanical Technology: An eight bit signed integer value representing the changes in encoder counts which have been accumulated since the last time a change in the encoder was read. Clockwise is positive. while count overflow: max value	Due to IRQ or need from master.	Display Unit controls

				shall be 127 for CW and -128 for CCW respectively Provider is FIDM & consumer is FIDM-Controller.		
0x0A	1	Debounced Switch Press ---Status(audio)	Read	Bit Mapped Switch States - 0: Seek Down - 1: Seek Up - 2: Menu/Enter - 3: Home - 4: Back - 5: Reserved - 6: Reserved - 7: Reserved Note: Once indicated status goes high it should not be cleared until read. Provider is FIDM & consumer is FIDM-Controller.	Due to IRQ or need from master.	Display Unit controls (bits 0-4) Reserved (bits5-7)
0x0B	1	Debounced input Status	Read	Bit Mapped Switch States - Bit 0: Power/Mute Button - Bit 1: :Bit7 TBD Note: Once indicated	Due to IRQ or need from master.	Display Unit controls

				status goes high it should not be cleared until read. Provider is FIDM & consumer is FIDM-Controller.		
0X0C	2	HVAC_White_General_BL_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> Backlighting Off.0xFFFF -> Backlighting at Maximum Level.LSB will be stored at the lower address.The ICS CTS will define maximum value.Provider is FIDM-Controller & consumer is FIDM	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms	Display Unit controls
0X0E	2	HVAC_Red_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> lighting Off.0xFFFF -> lighting at Maximum Level.LSB	Master must update on change with a maximum latency of 10ms. If no changes Master must	Display Unit controls

				will be stored at the lower address.The ICS CTS will define maximum value.Provider is FIDM-Controller & consumer is FIDM	update at a rate of 200ms seconds	
0X10	2	HVAC_Blue_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> lighting Off.0xFFFF -> lighting at Maximum Level.LSB will be stored at the lower address.The ICS CTS will define maximum value.Provider is FIDM-Controller & consumer is FIDM	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms	Display Unit controls
0X12	2	HVAC_Amber_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> lighting Off.0xFFFF -> lighting at	Master must update on change with a maximum latency of 10ms. If no changes	Display Unit controls

				Maximum Level.LSB will be stored at the lower address.The ICS CTS will define maximum value.Provider is FIDM-Controller & consumer is FIDM	Master must update at a rate of 200ms	
0X14	2	HVAC_AUX1_SwitchPack_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> Backlighting Off.0xFFFF ->Backlighting at Maximum Level.LSB will be stored at the lower address.The ICS CTS will define maximum value.Provider is FIDM-Controller & consumer is FIDM	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms	Display Unit controls
0x15	1	Front_HVAC_Indicator_Controls	Write	Bit0:Indicator Control AUTO [0->LED OFF;1->LED ON] Bit1:Indicator Control ON/OFF [0-	Master must update at a rate of 500ms	Display Unit controls

				>LED OFF;1->LED ON] Bit2:Indicator Control Front Def [0->LED OFF;1->LED ON] Bit3:Indicator Control Rear Def [0->LED OFF;1->LED ON] Bit4:Indicator Control A/C [0->LED OFF;1->LED ON] Bit5:Indicator Control HEAT [0->LED OFF;1->LED ON] Bit6-7:Reserved		
0x16	6	Diagnostic Status 1 Provider is FIDM & consumer is FIDM-Controller.	Read	**Byte 0:** 8-bit CRC value computed over next 5 bytes of this message **Byte 1:** Touch Controller ID. This is the same value as message 0x4, but must be sent within a CRC-protected	Periodic Read by Master Loss of communication shall be interpreted as no response to Master read requests for > 1000 ms.	Display Unit controls

				message. This is necessary because the Touch Diagnostic message format may be different between different touch controllers, and to achieve overall safety certification the FIDM controller must always decode the correct Touch Diagnostic message format.		
				Byte 2: DisplayID: This is the same value sent in 0x05, but must be sent within a CRC protected message for functional safety integrity.		
				Byte 3: Status bitfield Bit0: shall indicate OLDI/eDP error Bit1: shall indicate Display error Bit2: Shall		

				indicate Display Temporary ShutOff Bit3: shall indicate Gate/Source Driver error Bit4: shall indicate watchdog & or MCU software reset error Bit5: shall indicate abnormal power-on reset Bit6: shall indicate power managemen t IC error Bit7: shall indicate LVDS link lock error This message shall be read periodically at a 500ms rate driven by the FIDM Controller. U pon detection of any error on Bits 4, 5, 6, or 7, the FIDM-Controller shall re-initialize the Deserializer & Touch of FIDM Display module as defined in		

				FG.02.07.02 section 2.5.11.8. Note: Once an indicated status bit transitions to High, the FIDM shall not clear the bit (e.g. if the condition has been resolved), until after the message has been read by the FIDM Controller.		
				Byte 4: Status bitfield reserved for future expansion of additional error flags.(Default unused value shall be defined 0x00)		
				Byte 5: Counter value as follows: Upper 4-bit nibble always = 0xF Lower 4-bit nibble incremented by 1 on each transmission starting from 0x0 to 0xE and wrapping to 0x0 (0xF is		

				skipped).		
0x1C	4	Diagnostic Status 2 Provider is FIDM & consumer is FIDM-Controller.	Read	**Byte 0:** 8-bit CRC value computed over next 3 bytes of this message **Byte 1:** DisplayID: This is the same value sent in 0x05, but must be sent within a CRC protected message for functional safety integrity. **Byte 2:** Status bitfield with following bit mask definitions: Bit0 shall indicate Backlight error Bit1 shall indicate Backlight LED fault Bit2 shall indicate Backlight over voltage Bit3 shall indicate Backlight under voltage Bit4 shall indicate Backlight over	Periodic Read by Master Loss of communication shall be interpreted as no response to Master read requests for > 1000 ms.	Display Unit controls

				current Bit5 shall indicate Backlight temperature high Bit6 shall indicate Backlight temperature low Bit7 shall indicate Backlight error occurred in the past This message shall be read periodically at a 500ms rate driven by the FIDM Controller. Upon detection of any error the FIDM-Controller shall set a corresponding Diagnostic Trouble Code against the FIDM according to FG.02.07.02 section 2.5.11.9. Note: Once an indicated status bit transitions to High, the FIDM shall not clear the bit (e.g. if the condition		

				has been resolved), until after the message has been read by the FIDM Controller. **Byte 3:** Counter value as follows: Upper 4-bit nibble always = 0xF Lower 4-bit nibble incremented by 1 on each transmission starting from 0x0 to 0xE and wrapping to 0x0 (0xF is skipped).		
0x20	5	Display Backlight PWM Command	Write	**Byte 0:** 8-bit CRC value computed over next 4 bytes of this message **Bytes 1:** TargetDisplayID: ID of the target display sent for functional safety integrity purposes. This value must match the displays ID as reported by 0x05 or the	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms seconds.	Display Unit controls

				message is invalid. See rqmnts in FG.02.07.03.		
				Bytes 2 & 3: Commanded backlight PWM. Range is 0x0000 to 0xFFFF. 0x0000 -> Backlighting Off. 0xFFFF -> Backlighting at Maximum Level. LSB will be stored at the lower address. The Display CTS will define the resolution. Provider is FIDM-Controller & consumer is FIDM.		
				Byte 4: Counter value as follows: Upper 4-bit nibble always = 0xF Lower 4-bit nibble incremented by 1 on each transmission starting from 0x0 to 0xE and wrapping to 0x0 (0xF is		

				skipped).		
0x25	2	SwitchPack_Backlighting_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> Backlighting Off. 0xFFFF -> Backlighting at Maximum Level. LSB will be stored at the lower address. The ICS CTS will define the maximum value. Provider is FIDM-Controller & consumer is FIDM.	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms seconds	Display Unit controls Switch Pack Backlight PWM
0x27	2	Aux1_SwitchPack_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> Backlighting Off. 0xFFFF -> Backlighting at Maximum Level. LSB will be stored at the lower address. The ICS CTS will define maximum	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms seconds	Display Unit controls.

				value. Provider is FIDM-Controller & consumer is FIDM.		
0x29	2	Aux2_SwitchPack_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> Backlighting Off. 0xFFFF -> Backlighting at Maximum Level. LSB will be stored at the lower address. The ICS CTS will define maximum value. Provider is FIDM-Controller & consumer is FIDM.	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms-seconds	Display Unit controls
0x2B	2	Aux3_SwitchPack_PWM	Write	Commanded PWM Range is 0x0000 to 0xFFFF. 0x0000 -> Backlighting Off. 0xFFFF -> Backlighting at Maximum Level. LSB will be	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 200ms seconds	ICS Controls

				stored at the lower address. The ICS CTS will define maximum value. Provider is FIDM-Controller & consumer is FIDM.		
0x2D	2	Reset Counter	Read	Number of Resets of Display uC	On need from master	Display Unit controls
0x2F	2	Power Up Counter	Read	Number of power ups of Display uC	On need from master	Display Unit controls
0x31	1	Force Reset	Write	A reset will be performed based on the bit indicated: - Bit 0: Touch Controller - Bit 1: Display uC Bit2 to Bit7 is RESERVED	On need from master	Display Unit controls
0x34	2	Display Program Update	Write	Three Writes of two bytes each to enter BL mode within 10 milliseconds First write: 0x55AA (Access Key 0) Second write: 0x33CC (Access Key 1) Third write: any non-zero value (Session Key)	On need from master	Display Unit controls

0x36	5	PWM Multipliers	Read/Write	Specific implementation to be defined in CTRS		Reserved
0x50	1	Display_ModeControl_1 *Only supported by front passenger displays supporting a dynamic polarizer. Only supported by the FPI module and **not supported by the VCU**.	Write	Bit mapped: Bit0: Privacy Control 0-Privacy Disabled 1-Privacy Enabled. Privacy shall default to be disabled at every power on Bit1-Bit7: Reserved	Master must update on change with a maximum latency of 10ms. If no changes Master must update at a rate of 5 seconds	Display Unit controls
0x55	1	Touch Rotate	Read	Bit mapped: 0XA5- would denote rotate 0X5A- would not denote rotate Specific implementation to be defined in Display CTRS		Reserved
0x56	1	FIDM_Dim_Level_Request	Read	Display should report current scaled brightness (0-100%) [100%-->xFF ; 0% →0x00] The FIDM-Controller/RADIO uses this	On need from master	

				message to request the current display Dim level. This can be done for any reason. The Tier-1 will use a DID to read this out Specific brightness max value to be defined in Display CTS		
0x60	8	Software Part #	Read	8 Digit Part number, ASCII encoded, least significant digit at lowest address.	On need from master	Display Unit controls
0x68	8	Hardware Part #	Read	8 Digit Part number, ASCII encoded, least significant digit at lowest address.	On need from master	Display Unit controls

Note: data addresses in above table1 is still work in progress by FIDM supplier, FIDM-Controller must check with GM at the time of actual implementation.
The below signal definitions shall be applicable only for Physical HVAC implementation that needs to be supported by FIDM over LVDS.
The below table3 defines the interface between FIDM and FIDM controller for Physical HVAC connected to FIDM.

Data Address	**Len**	**Signal**	**Type**	**Rate**	**Range**	**Conversion**
0XD4	1	Bit0:FIDM Request Front HVAC Auto Setting Change	BLN	1000ms periodic read by Master & on 20ms event		$0 = FALSE $1 = TRUE

	3	Bit1-3:FIDM Request Front HVAC Auto Sensed State	ENM	trigger by FIDM. Periodic read at 1000ms rate driven by the Master(FIDM Controller)		$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	1	Bit4:FIDM Request Rear Defog Setting Change	BLN			$0 = FALSE $1 = TRUE
	3	Bit5-7:FIDM Request Rear Defog Sensed State	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	1	Bit8:FIDM Request Max Defrost Change	BLN			$0 = FALSE $1 = TRUE
	3	Bit9-11:FIDM Request Front Defrost Sensed State	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH

	1	Bit12:FIDM Request Recirculation Setting Change	BLN			$0 = FALSE $1 = TRUE
	3	Bit13-15:FIDM Request Recirculation Sensed State	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	1	Bit16:FIDM Request Compressor Setting Change	BLN			$0 = FALSE $1 = TRUE
	3	Bit17-19:FIDM Request Compressor Sensed State	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	1	Bit20:FIDM Request HEAT Setting Change	BLN			$0 = FALSE $1 = TRUE
	3	Bit21-23:FIDM Request HEAT Sensed	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED

		State				$3 = INDETERMINATE $4 = OORL $5 = OORH
	2	Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for above

0XD9	5	Bit0-4:FIDM Request Blower Level Change Front	SNM	1000ms periodic read by Master & on 20ms event trigger by FIDM. Periodic read at 1000ms rate driven by the Master(FIDM Controller)	-16.0 - 15.0 Counts	E = N * 1.0
	3	Bit5-7:FIDM Request Blower Level Front Sensed State 1	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	3	Bit8-10:FIDM Request Blower Level Front Sensed State 2	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	2	Bit11-12:FIDM	ENM			$0 = INDETERMINAT

		Request Blower Level Front Correlation Status				E $1 = PASS CRITERIA MET $2 = FAIL CRITERIA MET
	5	Bit13-17:FIDM Request Temperature Change Front Right	SNM		-16.0 - 15.0 Counts	E = N * 1.0
	3	Bit18-20:FIDM Request Temperature Front Right Sensed State 1	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	3	Bit21-23:FIDM Request Temperature Front Right Sensed State 2	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	2	Bit24-25:FIDM Request Temperature Front Right Correlation Status	ENM			$0 = INDETERMINATE $1 = PASS CRITERIA MET $2 = FAIL CRITERIA MET
	5	Bit26-	SNM		-16.0 - 15.0	E = N * 1.0

		30:FIDM Request Temperature Change Front Left			Counts	
	3	bit31-33:FIDM Request Temperature Front Left Sensed State 1	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	3	bit34-36:FIDM Request Temperature Front Left Sensed State 2	ENM			$0 = UNINITIALIZED $1 = RELEASED $2 = PRESSED $3 = INDETERMINATE $4 = OORL $5 = OORH
	2	bit37-38:FIDM Request Temperature Front Left Correlation Status	ENM			$0 = INDETERMINATE $1 = PASS CRITERIA MET $2 = FAIL CRITERIA MET
	2	bit39-40:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for above

0XE0	12	bit0-11:FIDM Request Front Blower Circuit 1 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	bit12-23:FIDM Request Front Blower Circuit 1 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	bit24-35:FIDM Request Front Blower Circuit 1 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	bit36-47:FIDM Request Front Blower Circuit 1 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit48-49:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for above

0XE8	12	bit0-11:FIDM Request Front Blower Circuit 2 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	bit12-23:FIDM Request Front Blower Circuit 2 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	bit24-35:FIDM Request Front Blower Circuit 2 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	bit36-47:FIDM Request Front Blower Circuit 2 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit48-49:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for above

0XF0	12	bit0-9:FIDM Request Front Right Temperature Circuit 1 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	bit10-19:FIDM Request Front Right Temperature Circuit 1 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	bit20-29:FIDM Request Front Right Temperature Circuit 1 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	bit30-39:FIDM Request Front Right Temperature Circuit 1 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit39-40:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for

						above

0XF8	12	FIDM Request Front Right Temperature Circuit 2 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Right Temperature Circuit 2 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Right Temperature Circuit 2 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Right Temperature Circuit 2 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit39-40:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for

						above

0X100	12	FIDM Request Front Right Temperature Circuit 3 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Right Temperature Circuit 3 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Right Temperature Circuit 3 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Right Temperature Circuit 3 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit39-40:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for

						above

0X108	12	FIDM Request Front Left Temperature Circuit 1 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 1 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 1 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 1 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit39-40:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for

						above

0X110	12	FIDM Request Front Left Temperature Circuit 2 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 2 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 2 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 2 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit39-40:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for

						above

0X118	12	FIDM Request Front Left Temperature Circuit 3 Total Evaluations	UNM	1000ms periodic read by Master. Periodic read at 1000ms rate driven by the FIDM Controller	0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 3 Out Of Range High Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 3 Out Of Range Low Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	12	FIDM Request Front Left Temperature Circuit 3 Indeterminate Fail Evaluations	UNM		0.0 - 4095.0Counts	E = N * 1.0
	2	bit39-40:Alive Rolling Count	UNM		0-3 counts	incremented by 1 for each transmission
	8	CRC	UNM			8-bit CRC for

						above

1.2.2 FIDM Touch Controller Messages

This section defines the messages from the FIDM's internal touch controller. The touch controller is a physically separate device from the FIDM's microprocessor and has a separate dedicated I2C address.

Note that the touch controller driver, implemented within the FIDM Controller, will define the exact I2C message byte format and ordering. This section defines the sequence of parameters that must be delivered from the touch controller driver.

1.2.2.1 Touch Event Message

The Touch Event Message is defined as follows:

1. 12-bit XTouchCoordinate: The X coordinate of the touch point

2. 12-bit YCoordinate: The Y coordinate of the touch point

1.2.2.2 Periodic Touch Panel Diagnostic Message

The Touch Panel Diagnostic Status message is defined in this section.

The Touch Panel Diagnostic Status message is intended to be read by the FIDM Controller and triggered periodically via the Touch Controller GPIO interrupt line at a constant rate of 200 ms

A timeout of the Touch Controller shall be defined as 1000 ms without a response from the Touch Controller.

The Touch Panel Diagnostic Status message is defined as follows.

Byte Number	Parameter	Description
0	CRCValue	8-bit CRC value computed per CRC8 algorithm defined in section 1.2.3 of this specification as follows: The CRC is computed across message bytes 1 through 40.
1.bit0	BulkFailStatus	Boolean flag where True indicates a bulk failure of the touch screen. "bulk failure" is defined within the FIDM CTRS.
1.bit1	LineFailStatus	Boolean flag where: True indicates one or more line failures are present in the touch screen. RowLineFailArray and ColumnLineFailArray contain the specific line failures. False indicates no line failures detected.
2..20	RowLineFailArray	19-byte array packed bitfield of row line. Each bit is a boolean flag corresponding to the fault status of each touch trace row line. True = Fault, False

		= No Fault Row line flags are packed LSBit first as follows: Byte0.bit0 -> row 0, Byte0.bit1 -> row 1, ... Byte18.bit7 -> maximum row index of (19*8)-1 = row 151.
21.39	ColumnLineFailArray	19-byte array packed bitfield of column line. Each bit is a boolean flag corresponding to the fault status of each touch trace column line. True = Fault, False = No Fault Column line flags are packed LSBit first as follows: Byte0.bit0 -> column 0, Byte0.bit1 -> column 1, ... Byte18.bit7 -> maximum column index of (19*8)-1 = column 151.
40	Counter	Counter value as follows: Upper 4-bit nibble always = 0xF Lower 4-bit nibble incremented by 1 on each transmission to the FIDM-Controller, starting from 0x0 to 0xE and wrapping to 0x0 (0xF is skipped)

1.2.2.3 Knob on Display Event

The Knob On Display Event is defined as follows

Accumulator Counts Change +/-127(8-bit signed integer)	Diagnostic (fail/no fail) (1-bit)	Rolling counter (16-bit)

An eight-bit signed integer value representing the changes in encoder counts which have been accumulated since the last time a change in the knob was read. Clockwise is positive. Reset the accumulator buffer after count overflow.

1.2.2.4 Knob on Display Diagnostic

The Knob On Display Diagnostic shall be sent periodically by the FIDM microprocessor at a background rate of 500ms when no failure is present. Upon detection of a failure, the rate can increase to 20 ms. This message does not require a CRC value.

Diagnostic Data	
Diagnostic fail(1-bit)	Rolling Counter (16-bit)

Note: Knob related information is work in progress, check with GM at the time of implementation.

1.2.3 CRC8 Computation Requirements

Specific diagnostic status messages from the FIDM microprocessor and FIDM touch controller as well as the brightness command message are protected with an 8-bit CRC in order to meet functional safety goals. The specific messages for which CRC protection is required are defined in the message table within this module.

This section defines the CRC8 algorithm to be used to construct and verify all protected messages.

The 8-bit CRC for all protected messages within this specification shall be computed by the FIDM as follows:

1. Polynomial is from CRC8 SAE J1850: 0x1D (x8 + x4 + x3 + x2 +1).

2. Initial value = 0x00

3. XOR value = 0x00 (value XORed into the final register before returning the result CRC value)

1.2.4 Display Brightness Command CRC8 Computation

The 16-bit brightness value sent in the 0x20 Display Brightness Command message via the I2C interface must be protected with a CRC8 value in order to satisfy functional safety and regulatory data integrity requirements. The FIDM Controller generates the CRC8 protection value according to the algorithm defined above.

The FIDM Microprocessor verifies the received code and handles verification errors as defined within the *FG.02.07.03 Fully Integrated Display Module* requirements module.

1.3 FPDLink Lock Status

Display depends on the dynamic LOCK status to monitor the FPDLink cable plug/unplug connection and to recovery the display or other controls.

984 and 988 Deserializers support single or dual FPD ports, some GPIO pins are defaulted to output a dynamic LOCK status for each port as shown in below table.

984/988 Deserializer GPIOs	FPD Port#	Remarks
GPIO 4	0	FPD Port 0 LOCK status
GPIO 5	1	FPD Port1 LOCK status
GPIO 6	0&1	AND of FPD Port 0 and Port 1 LOCK status

FIDM Controller FG.02.07.02 implements the configuration of linking the LOCK signaling from Serializer to the assigned GPIO at Deserializer as shown in the table. The GPIO# is not needed to match to each other

FIDM FG.02.07.03 no need a configuration to bring up the LOCK from the designated GPIO at Deserializer. The not used GPIO can be configured for a general I/O purpose.

1.4 FPDLink AC Grounding

The ground isolation of the FPDLink INPUT and OUTPUT connector is important to prevent a condition where current can flow between downstream devices and the source device which can cause some potential problems. The device must pass EMC test.
OUTPUT connector locates at video source device;
INPUT connector locates at video sink device (or downstream device);
In general, shorting OUTPUT connector GND pins to main ground on the “source” side of the FPDLink, and employing an AC grounding strategy on INPUT connector on the “ Sink” side of the FPDLink.

1.5 Displays Identification

The table below defines the master list of variants identified by a Display ID

DISPLAY GROUP	DISPLAY_ID	DESCRIPTION
IPD = 0	0	8” SLIM SD I_I
	2	11” SLIM SD I_I
	3	11" SLIM SD T_I
	13	CHEVY PERF_I
	18	11” CHEVY_T_30BIT
	19	11" GMC_A_30BIT
	29	11"_CHEVY_984_T_I_30bit
VCD = 1	20	11"_MM_I
	21	11" SLIM SD I_V
	24	11" SLIM SD T_V
	5	11+ SD I_V
	6	11+ SD T_V
	8	CAD FF1_V 34"
	9	BUICK FF_V 30"
	10	GMC 15"_V
	11	GMC 16.8"_V

	12	CHEVY FF_V 17.7"
	14	CAD FF2-1_V 35"
	25	CAD FF2-2_V 20"
	15	CHEVY PERF2_V
	22	11" MM_V
Fcc = 2	23	8" SLIM SD_TN_F
	16	10.3"_F
	17	CHEVY PERF3_F
	30	8"_SLIM_SD_T_FCC
REARHVAC = 3	27	8" SLIM SD_T_CC
	28	12.6"_RSI
	31	8"_SLIM_SD_T_R
HUD = 4(Please refer HUD Module for any additional details) (The HUD ids are separate and are not to be considered to be part of Display IDs)	0	1.5 X 3.5
	1	2 X 4
	2	2 X 4
	3	3 X 7
	4	3 X 7
	5	C-HUD
	6	DUAL PLANE

1.6 Touch Interface

This section defines the touch interface over the I2C back channel in a touch-screen capable system.

The table below defines the Touch Controller ID sent in both the 0x04 and 0x1A messages in the table above.

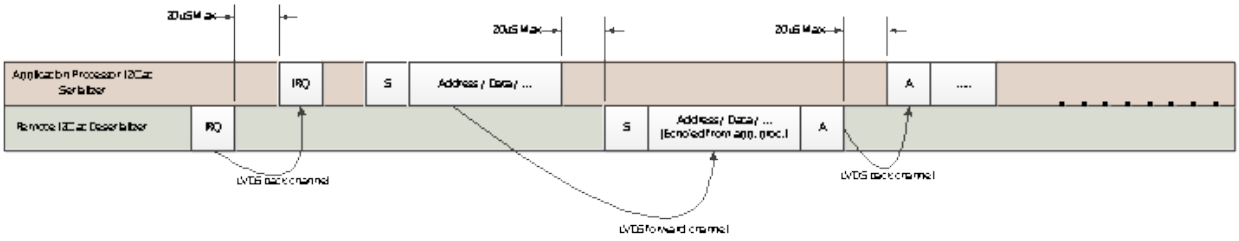
Identification Number	Supplier	Part Number	I2C Address

0	Microchip	mXTxxx	0x4B
1	Cypress	CYAT817xxx	0X24
2	Synaptics	TD7XXX	0x70
3	Himax	HX8xxx	0x48

The touch panel interface consists of I2C serial communications with an interrupt line over the LVDS forward/back channel. On the Serializer side the I2C touch panel interface will connect directly to the application processor. On the Deserializer side, the I2C interface will directly connect to the touch controller. The I2C interface can be clocked at up to 3.4MHz at local (Between application processor and Serializer) and 1MHz between at remote (Between Serializer and Desializer). The application processor I2C messages received at the Serializer will be transmitted to I2C touch controller via the Deserializer. The touch controller I2C messages and responses at the Deserializer will be transmitted to the application processor via the Serializer. An interrupt signal will also be transferred from the touch controller at the Deserializer to the application processor via the Serializer. The following timing will be a reference to design:

Timing Requirements (See diagram below)	
I2C Clock Rate	1 MHz
Time from touch controller interrupt received at the deserializer to interrupt sent to the application processor at the serializer.	20uS maximum
Time from end of application processor message at the serializer to the beginning of touch controller message at the deserializer.	20uS maximum
Time from the end of touch controller message at the deserializer to the beginning of application processor message the serializer.	20uS maximum

Note: The 20uS maximum time requirement above is internal feature/function of the Serializer/Deserializer. No special configuration is needed.



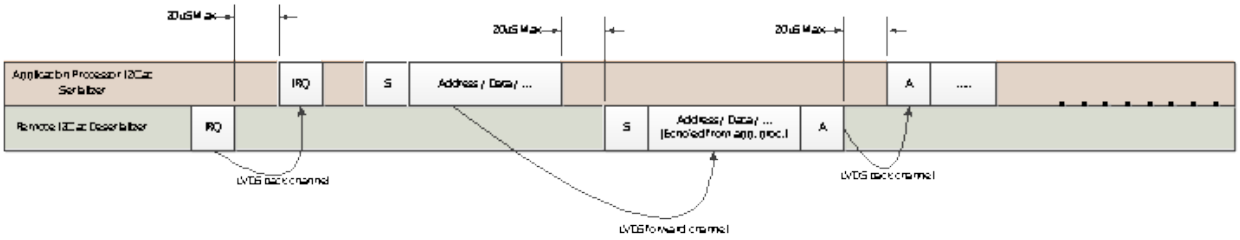
Refer to the I2C Message map for specific I2C message details.

1.6.1 FIDM Controls Interface

The FIDM based module (i.e. the Center Stack Module - ICS) Controls interface consist of I2C serial communications with an interrupt line over the LVDS forward/back channel. On the Serializer side, the I2C interface will connect directly to the application processor. On the Deserializer side, the I2C interface will directly connect to the FIDM Controls. The I2C interface can be clocked at up to 3.4MHz at local and 1MHz between at remote. The application processor I2C messages received at the serializer will be transmitted to I2C FIDM Controls via the Deserializer. The FIDM Controls I2C messages and responses at the Deserializer will be transmitted to the application processor via the Serializer. This I2C interface will be shared with the touch controller. The FIDM Controls will have a dedicated interrupt signal transferred from the FIDM Controls at the Deserializer to the application processor via the Serializer. The following timing will be reference to design:

Timing Requirements (See diagram below)	
I2C Clock Rate	1MHz
Time from FIDM Controls interrupt received at the deserializer to interrupt sent to the application processor at the serializer.	20uS maximum
Time from end of application processor message at the serializer to the beginning of FIDM Controls message at the deserializer.	20uS maximum
Time from the end of FIDM Controls message at the deserializer to the beginning of application processor message the deserializer.	20uS maximum

Note: The 20uS maximum time requirement above is internal feature/function of the Serializer/Deserializer. No special configuration is needed.



See I2C message map for additional details about the I2C interface.

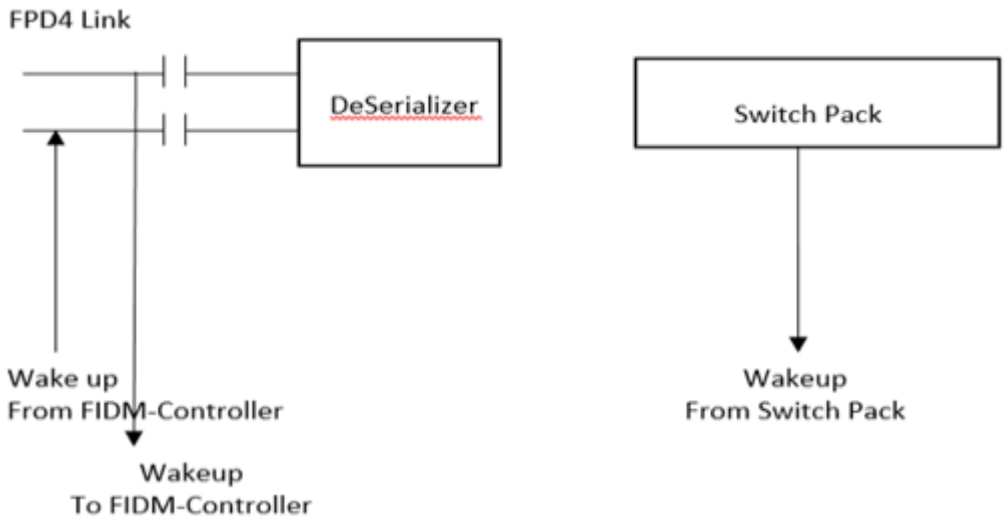
1.7 Wakeup and Power Moding

The FIDM (FG.02.07.03) and FIDM-Controller (FG.02.07.02) can wakeup either by applying a DC level on the LVDS interface pair or, for the FIDM itself, via a wakeup from an attached Switch Pack.

1) FPD Link LVDS Signaling – Applied to all variants

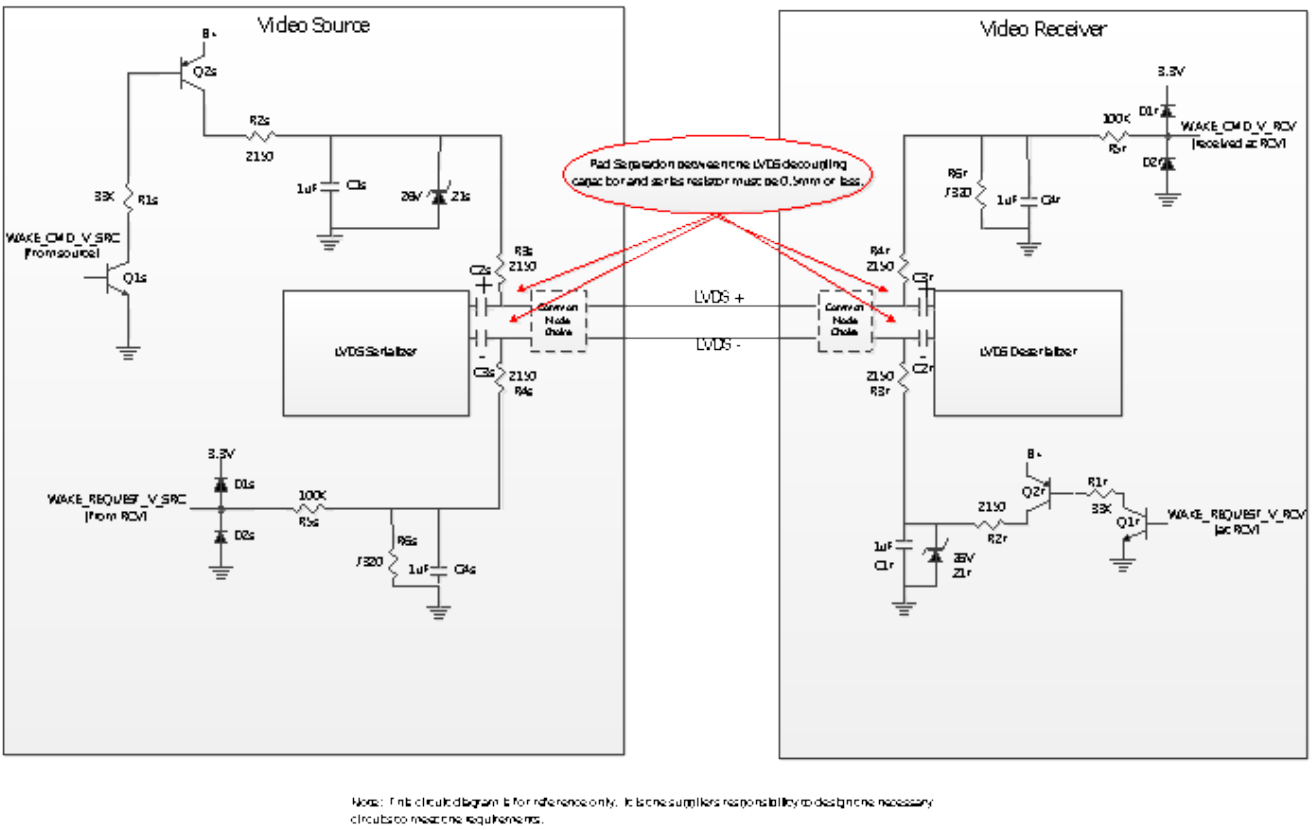
2) Local wakeup from switch pack power button – Applied to the Center Stack module and

other Large format variants



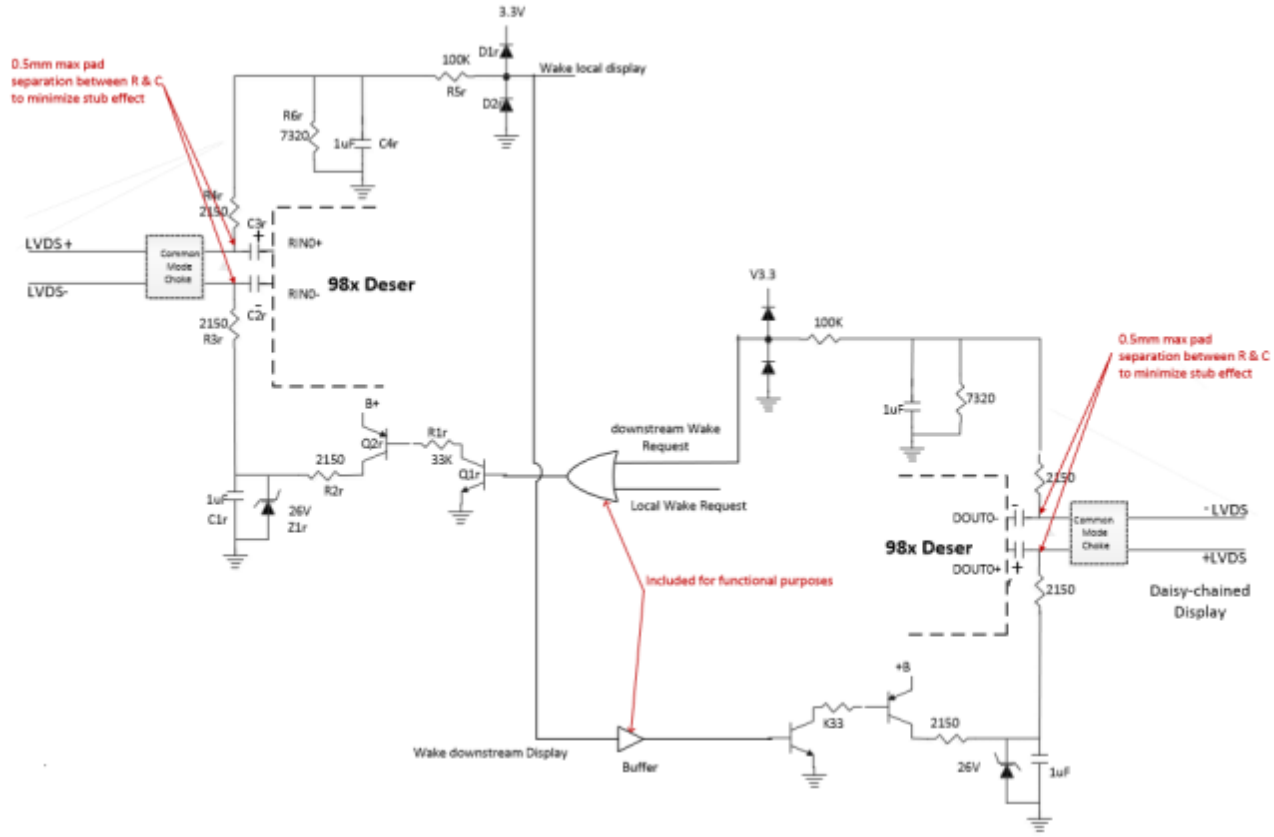
1.7.1 Wakeup via DC bias on LVDS signal pair

Both the FIDM and the FIDM controller can wake each other up via a DC bias level applied to the LVDS signal pair as defined within this section. The circuit below provides a reference circuit for the system implementation showing the required circuitry on each side.



Note: This circuit diagram is for reference only. It is the supplier's responsibility to design the necessary circuits to meet the requirements.

Daisy-chained Display Wakeup



Circuitry requirements:

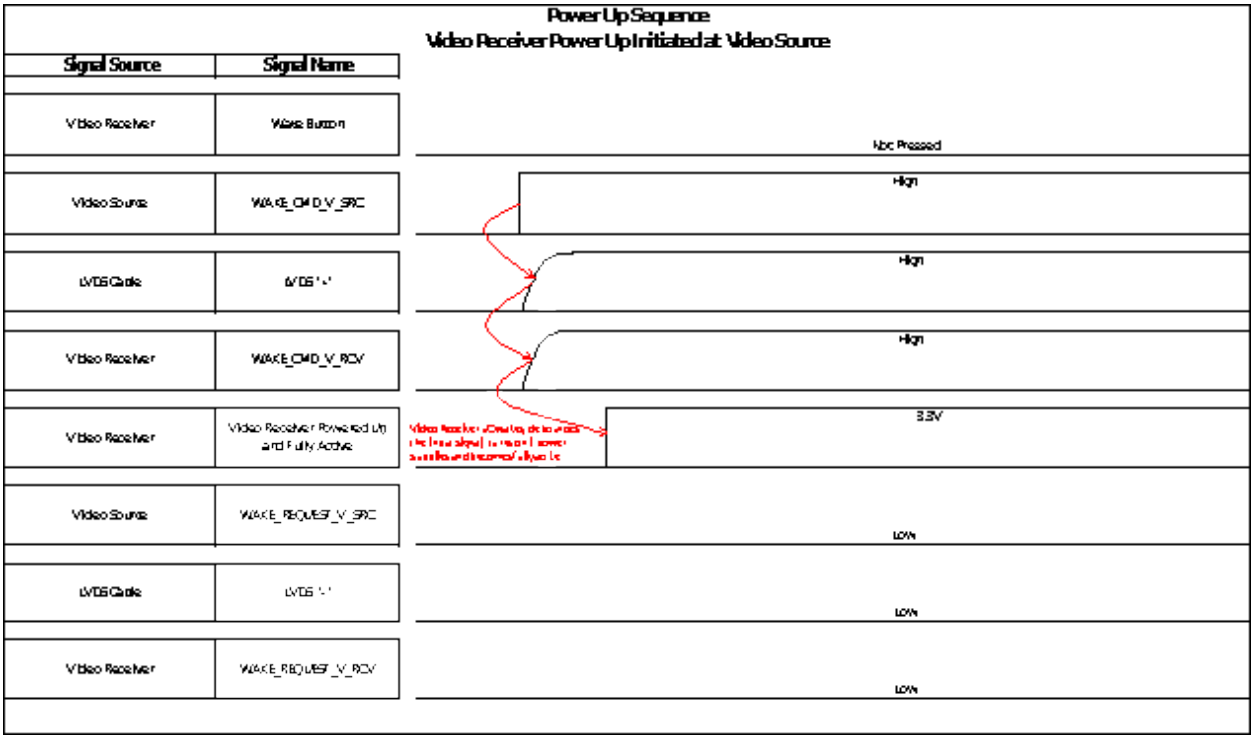
- Pad separation between the LVDS decoupling capacitor (C2x & C3x) and DC level sensing series resistor (R3x & R4x) must be 0.5mm or less.
- Nominal 100 Ω differential input characteristics of the LVDS (+ & -) input of the “Video Source” or “Video Receiver” above 32KHz must not be altered by more than 2% by the power moding output or input circuit.
- The power moding circuit must load each LVDS (+ & -) signal line with a 2.15K+/-3% resistance above 32KHz.
- Specific LVDS Input side requirements (LVDS + for Video Receiver & LVDS – for Video Source as shown on reference circuit):
  - A low pass filter limiting any input frequencies with a first order cut off frequency at 115Hz maximum as shown with a worst-case analysis. For reference see components R4x, R6x, & C4x.
  - Input circuits receiving WAKE_REQUEST or WAKE_CMD must be protected from input transients and voltage levels exceeding the input circuit requirements. For reference see components D1x, & D2x.
  - Must always detect a low when <= 1.26V is applied to the “Video Source” LVDS - or “Video Receiver” LVDS + input. The low signal on this node is achieved with R6 as shown in the reference diagram.
  - Must always detect a high when >= 3.17V is applied to “Video Source” LVDS -

or “Video Receiver” LVDS + input. See reference circuit above.

- Specific LVDS Output side requirements:
  - A low pass filter must limit any frequencies applied to the LVDS signals with a first order cut off frequency at 88.8Hz maximum as shown with a worst-case analysis. For reference see components R2x, & C1x.
  - LVDS signals must be protected from any voltage greater than 26V +/- 20%. For reference see components Z1x.
  - An output logic low condition must be a high impedance and source less than 1uA of current out of the “Video Source” LVDS + or “Video Receiver” LVDS - input.
  - When the output is logic high, the output must provide a high voltage >= 3.37V at the “Video Source” LVDS + or “Video Receiver” LVDS - output with a 9186 Ohm load on LVDS line.

Note: All voltage levels are average DC voltages.

The following timing diagrams showing typical power moding sequences with FPD Link LVDS signaling:

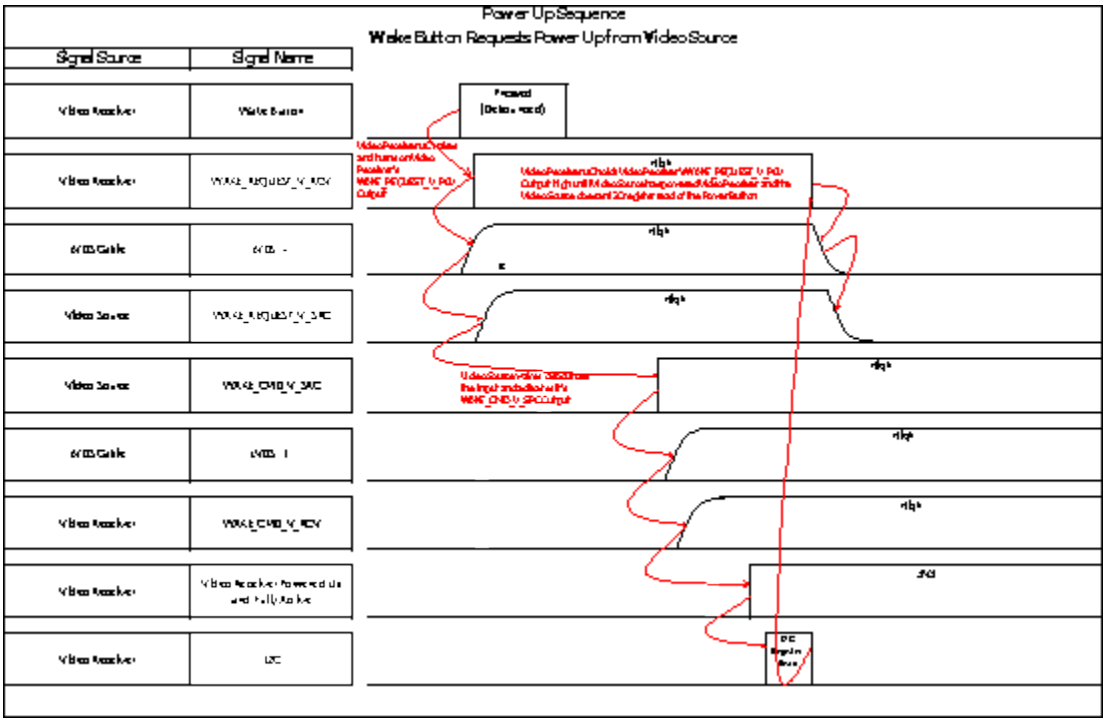


1.7.2 Power Up by Switch Pack Wakeup Button

Local wake up from switch pack power button will be used for the Centre Stack module and large format displays. The switch pack provides a discrete switch input for wake up. If the controller is already operational, then activation of this input will create a switch pack interrupt request on GPIO with interrupt/reporting behavior like other switches. If the controller is in the sleep state, activation will trigger the LVDS wakeup line to wake up the

controller.

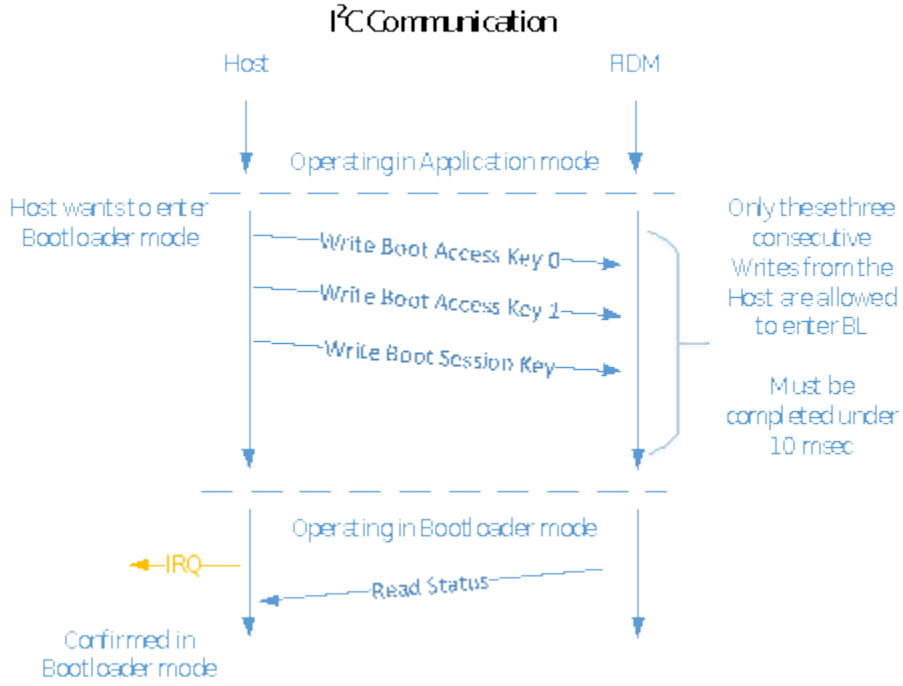
The following timing diagrams showing typical power moding sequences with switch pack wake up button:



1.8 FIDM Software Update requirements

This section describes the update of FIDM microprocessor software transferred from and controlled by the FIDM Controller (shown as “Host”).FIDM shall support to receive Touch software or any of internal FIDM sub-component software update as single update package from FIDM-Controller

Below is the I^2C sequence to enter FIDM Bootloader (BL) mode from the FIDM Application mode.



The Application Access Keys consist of four bytes that are previously stored in the application memory block of the FIDM. The specific Application Access Keys (four bytes) values must be known to the Host.

There are two bytes used for a Session Key that are temporary and only used during a session of the Bootloader (POR Reset clears the Session Key to zero). The host can choose any non-zero value for these however if the application is invalid (CRC32 does not match) and a POR occurs the Session Key is set to zero. The Session key is stored in SRAM of the FIDM and are used to validate critical host requests:

1. Unlock the flash pages used by the application,
2. Erase the application memory
3. Write a flash page (64 bytes).

All I²C transactions that are above address 0x80 must append a CRC to the write operation. When I²C reads are performed above address 0x80 a CRC is available at the end of the data. The 16-bit CCITT CRC (0x1021) polynomial is used for read and write transactions.

1.8.1 Example 16-bit CCITT CRC

Data is = 0x0001

CRC16 = 0x3331

		Data	Data	CRC	CRC	
START	SLA-R	0x01	0x00	0x31	0x33	STOP

1.8.2 Table of I2C Addresses available in FIDM Bootloader mode

The table2 below lists the available I2C address in the FIDM during the Bootloader mode.

Data Address	Data Length (Bytes)		Data	Data Direction (From Device)	Data Type
0x00[1)]	2		Hardware Version	Read	Hardware Version of the FIDM
0x02[1)]	2		Firmware Version	Read	Firmware Version of the FIDM
0x04[1)]	1		Touch Controller ID	Read	Identification number of touch controller used.
0x05[1)]	1		Display ID	Read	Identification number of display resolution / timing used.
0x20	5		Display Backlight Command	Write	Command backlight duty cycle. Refer Section table 1.2.1 above
0x31	1		Force Reset	Write	A reset will be performed based on the bit indicated: - Bit 0: Touch Controller - Bit 1: FIDM uC Refer section table 1.2.1 above
0x34	2		FIDM Program Update	Write/Read	Three Writes of two bytes each to enter BL mode within 10 milliseconds First write: 0x55AA (Access Key 0) Second write: 0x33CC (Access Key 1) Third write: any non-zero value (Session Key) Refer section table 1.2.1 above
0x80	4		Bootloader Status	Read	Provides Bootloader State Machine Status Byte0:Bootloader Status1 Bit Mapped States - 0: Bootloader (BL) Mode Active - 1: Application Unlocked - 2: Application Erased - 3: Flash Write Successful - 4: Session Key Error - 5: Erase Key Error - 6: CRC Error

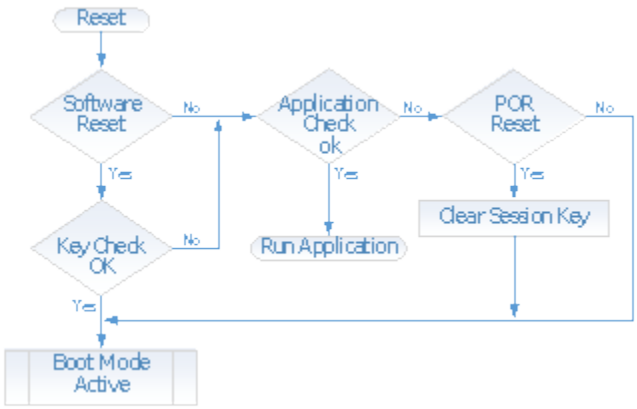
					- 7: Voltage Error Byte1:Bootloader Status2 - 0: Programming Error - 1: Unknown Error Byte2:CRC16 LSB Byte3:CRC16 MSB
0x84	4		Unlock Application Memory	Write	Change the lock bits on the application memory to allow write access
0x88	5		Erase Application Memory	Write	Erases all pages of the application memory (if first checks if there is data on the page and performs the erase operation if it is not already erased)
0x8D	71		Write Flash Page	Write	Write the flash page with the contents of the specified address

1)Invalid values will be reported if the application memory locations are in the erased state
Sticky Bits (only cleared on a Reset)

1.8.3 Overview of the FIDM process to enter bootloader mode

An overview of the FIDM process to enter bootloader mode. Requirements include:

- Boot Access Keys received and validated via the I^2C bus within 10 msec.
- Application writes the Session Keys to SRAM
- Application writes a BL key to the SRAM
- Application issues a software reset
- The startup code checks for a software reset and a valid BL key is in SRAM

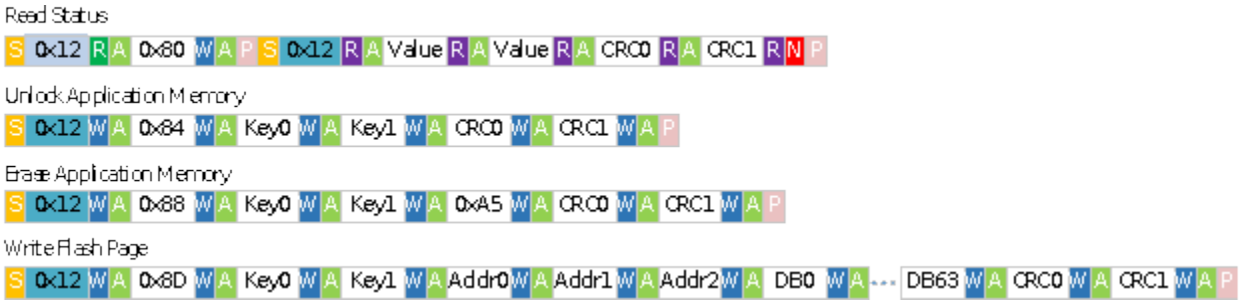


Bootloader mode is indicated by:

- BL mode should be indicated by FIDM controller, commanding a 95% duty cycle on the switch pack Backlighting.
- The IRQ line is asserted by the BL and the Host should read the BL status (0x80).

I²C Messages accepted by the FIDM bootloader are from 0x00 to 0x05, 0x20, 0x21, 0x40, and 0x80 to 0xD4.

1.8.4 Overview of the I2C message format for BL mode



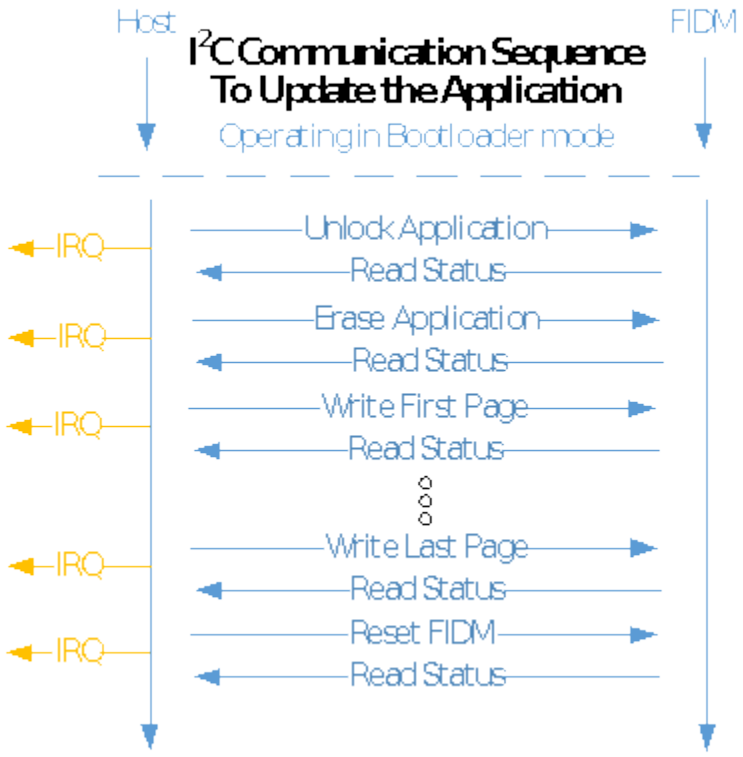
The 16-bit and 24-bit values are written/read MSB first (i.e. CRC0 is LSB of the CRC-16 and CRC1 is the MSB of the CRC-16).

The Data bytes written/read for the flash page correspond to the byte position as it would be in flash memory. A flash page on this device is 64 bytes long and the address sent by the host must be on 64 byte boundaries.

In addition, before an Erase Application Memory or Write Flash Page operation the Battery voltage is checked and it must be above 9V otherwise an error is flagged.

While a Flash operation is active I²C messages are not accepted by the FIDM and are lost.

The Bootloader mode will assert the IRQ line on any change to the BL Status register. The IRQ will be cleared once this register is read.



1.8.5 End to End Flow Overview

FIDM-Controller must be primary programming master (flash master) for all the display subsystems [AUX ICS HVAC IPC HUD].

FIDM-Controller must support Peripheral SW programming (AUX ICS HVAC IPC HUD) per GM USB/OTA requirements.

FIDM-Controller must act as Host for programming Display subsystems (AUX ICS HVAC IPC HUD) which, in this case is peripherals.

FIDM-Controller should include alerts to report the outcome of Display subsystem programming, these alerts should meet current operational signatures per GM back office requirements in case OTA (please refer OTA requirements) and After Sales Dealer service requirements for USB, accordingly.

1.8.6 FIDM-Controller side → update flow

The Display update from FIDM-Controller implementation standpoint can be broadly classified into following

1) Pre-update (pre-condition check)

2) During Update

3) Post-update

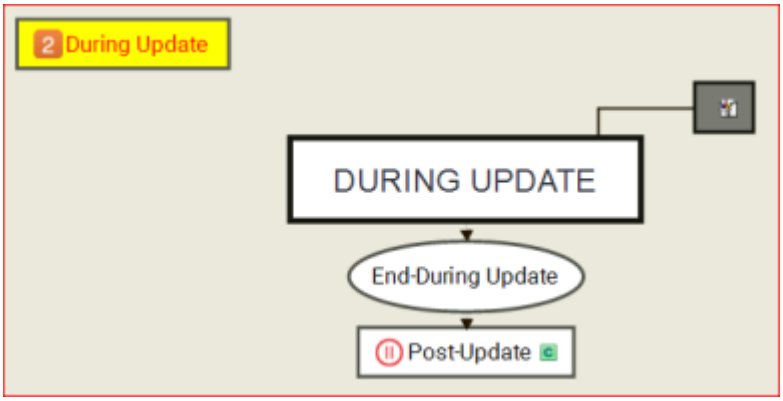
4) Functional Check after re-flash

1.8.7 Implementation Flow diagram

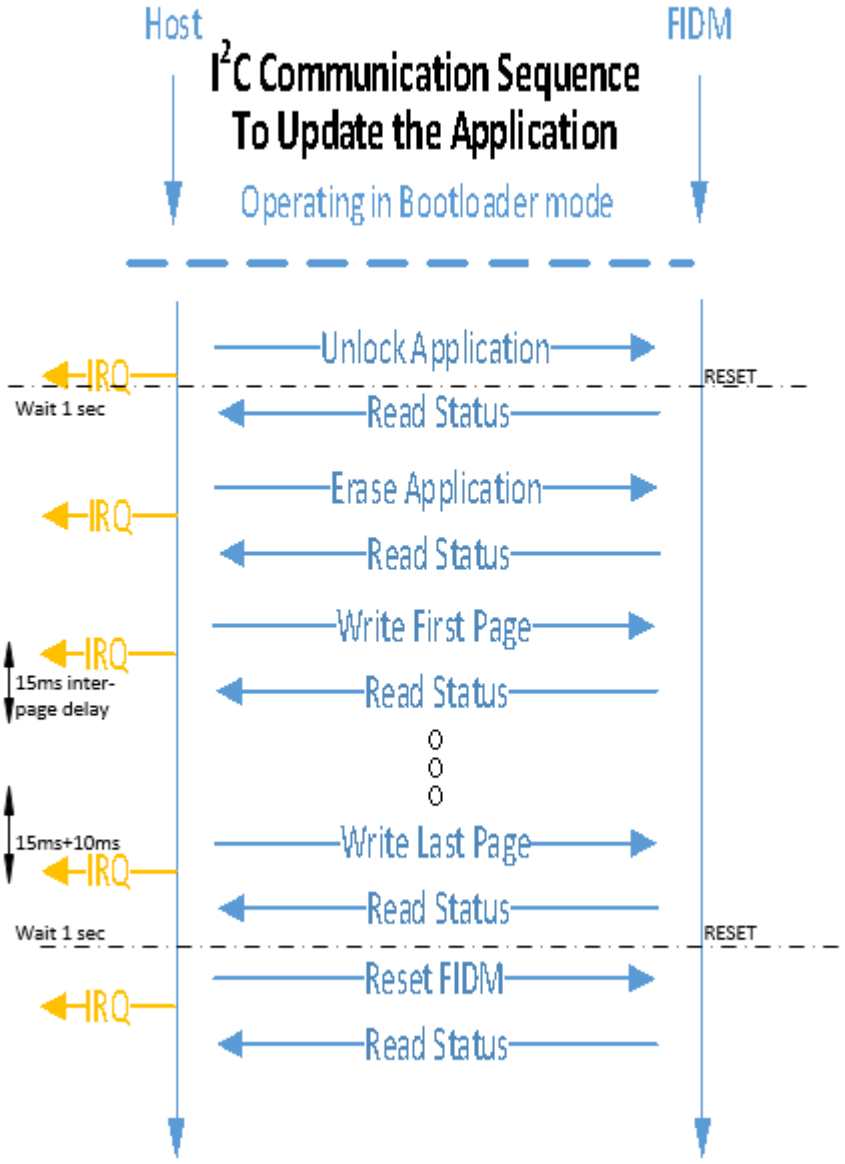
1.8.7.1 Stage-1 Pre-Update

1.8.7.2

1.8.7.3 Stage-2 During Update



Note- any failure in this phase, logic will again retry for 5 times

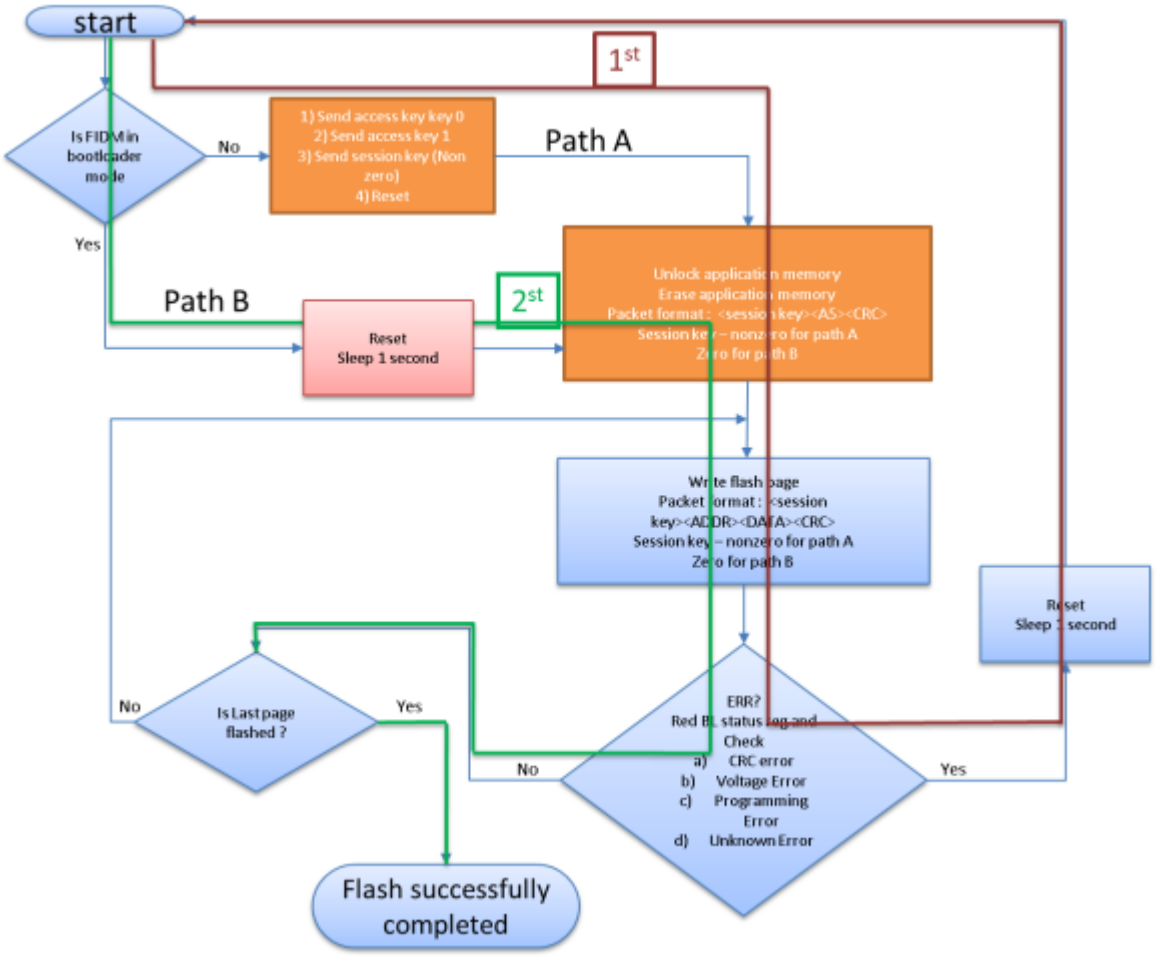


1.8.7.4 Stage-3 Post Update
1.8.7.5
1.8.7.6 Stage-4 Functional Check
1.8.7.7

1.8.8 Failure Scenarios

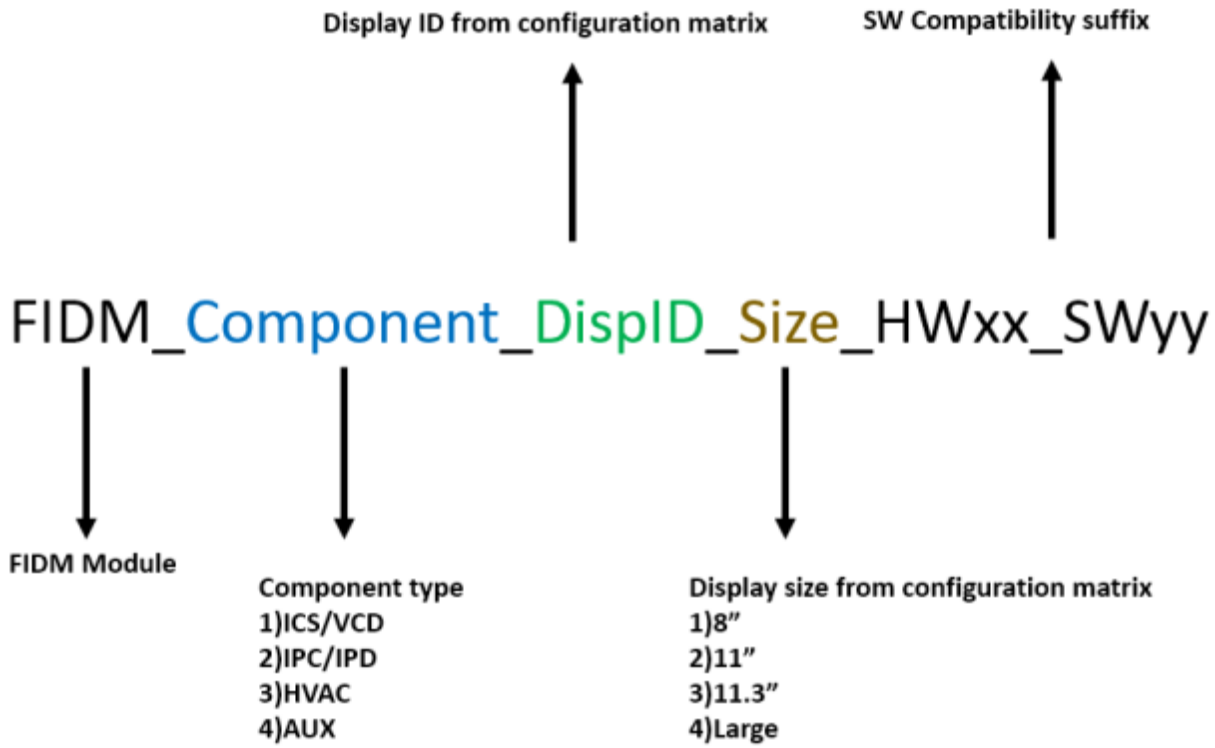
1. Power Interruption
2. FPD Data Link interruption
3. Voltage Range check
4. Incorrect Display Variant SW detection on CSM/Radio side.

The flashing operation needs to be robust to recover from any of the above failure scenario.



1.8.9 File naming convention

FIDM-Controller would implement logic to parse the filename of update file and apply respective update if applicable for the Display that is being paired FIDM-Controller.



Note:

The Display supplier is responsible to ensure the integrity of the file naming and its association with the release cadence of corresponding AUX ICS HVAC IPC variant and FIDM-Controller release cadence. The compatibility and Part number configuration logistics would be handled by Display supplier in close co-ordination with FIDM-Controller logistics and GM back office.

1.8.10 FIDM Controller High-Level Application Layer Update Handling

FIDM Controller High-Level Application Layer Update Handling

The Requirement for FIDM update should be for FIDM-Controller to handle the Display update in the same lines as its internal sub-components. FIDM-Controller should implement a parsing logic to detect and flash corresponding interfacing Display module.

FIDM Controller should implement diagnostic routine to read SW and HW PN of AUX ICS HVAC IPC HUD.

8bytes	Hardware Part # AUX ICS HVAC IPC HUD	Read	8 Digit Part number, ASCII encoded, least significant digit at lowest address	On need from master(FIDM-Controller)	
8bytes	Software Part # AUX ICS HVAC	Read	8 Digit Part number, ASCII encoded, least significant digit at lowest address	On need from master(FIDM-Controller)	

	IPC HUD				

Note- AUX ICS HVAC IPC HUD should enable support to read PN’s from both Bootloader & application

AUX ICS HVAC IPC HUD may have separate SW module id(SWMI), DD table item and manifest entry appropriately, only if required by FIDM-Controller, to meet reflash guidelines for Display Subsystem

#1_HUD	#2_AUX	#3_IPC	#4_ICS	#5_HVAC
SWPN1	SWPN2	SWPN3	SWPN4	SWPN5
HWPN1	HWPN2	HWPN3	HWPN4	HWPN5

Note: Throughout all FIDM documents following terms are used interchangeably, please check with GM-Display if unclear

1)IPD→IPC

2)VCD→ICS

1.8.11 Update Diagnostics

The following DIDs will be used:

- DID $ AUX ICS HVAC IPC HUD - Software Part Number
- DID $ AUX ICS HVAC IPC HUD - Hardware Part Number

Refer to FIDM-Controller (FG.02.07.02) for diagnostics Spec. for DID.